Chapter 5 — Microprocessor and Microcontroller

RRB JE CBT-2 · Electronics & Allied Engineering (CEN 05/2025, Annexure-VII C) · Syllabus Unit 5

> **Unit coverage:** Introduction to microprocessor · 8085 microprocessor working · Assembly Language programming · Peripherals & other microprocessors · Microcontrollers.

📊 Weightage & Trend Box

Parameter	Detail
Approx. weight	**~8% of technical marks (≈ 4 out of 50 questions)** in the 22-04-2025 CBT-2 paper
Difficulty	Easy–Moderate; mostly single-concept, formula/fact recall + one small program trace
Best scoring areas	8085 program tracing, interrupt facts, 8051 basic architecture, DMA handshake

Questions actually observed in the 22-04-2025 CBT-2 paper (solved fully in the PYQ Corner):

1. **8085 program trace:** `MVI B, 06H` → `MVI A, F2H` → `ADD B` — what is the accumulator content? *(Answer: F2H + 06H = **F8H**)*
2. **8051 register banks:** How many register banks does the 8051 have? *(Answer: **four**, each with R0–R7)*
3. **8085 interrupts:** Which interrupt is **positive-edge-triggered**? *(Answer: **RST 7.5**)*
4. **DMA:** When is the **DACK** (DMA acknowledge) signal issued in a DMA transfer? *(Answer: after the CPU grants the bus via HLDA and the DMA controller places the address — DACK tells the peripheral the transfer cycle is starting.)*

Trend takeaways

- Every recent paper has **one small 8085 program-tracing question** (usually MVI/ADD/SUB with flag or accumulator result). Practise hex arithmetic until it is automatic.
- Interrupt facts (edge vs level trigger, vector addresses, TRAP highest priority) and 8051 basics (128 B RAM, 4 register banks, 5 interrupts, 12 clocks per machine cycle) are recycled almost verbatim.
- Peripheral chips are tested at the "one-line function" level: 8255 = PPI, 8257 = DMA, 8259 = PIC, 8279 = keyboard/display — plus one deeper question (control word or handshake) occasionally.

5.1 Introduction to Microprocessors

5.1.1 What is a microprocessor?

A **microprocessor (µP)** is a programmable, clock-driven, register-based electronic device fabricated on a single chip (VLSI) that:

1. reads binary **instructions** from memory,
2. **decodes** them, and
3. **executes** them — performing arithmetic/logic operations on data and producing results/control actions.

A microprocessor by itself is only a CPU. To make a working computer it needs **memory** (ROM + RAM), **I/O devices**, and interconnecting **buses**. The combination is a **microcomputer**:

```
Microcomputer = Microprocessor (CPU) + Memory + I/O + Buses
```

5.1.2 Basic terms you must know

Term	Meaning	8085 value
Word length	Number of bits the µP processes at a time (data bus width)	8 bits → "8-bit µP"
Address bus	Unidirectional bus carrying memory/I-O addresses	16 lines → 2^{16} = **64 KB** address space
Data bus	Bidirectional bus carrying data	8 lines
Control bus	Individual lines carrying timing/control signals (RD, WR, ALE …)	—
Clock	Fixed-frequency square wave synchronising all operations	Crystal 6.144 MHz; internal clock = crystal/2 = **3.072 MHz** (max rated internal clock of 8085A = 3 MHz class)
T-state	One period of the internal clock; smallest unit of processor timing	1/3.072 MHz ≈ 325.5 ns (with 6.144 MHz crystal)
Machine cycle	Time to complete one memory/I-O access (3–6 T-states)	e.g. opcode fetch = 4T
Instruction cycle	Time to fetch + execute one full instruction (1–5 machine cycles)	e.g. `MOV A,B` = 4T; `LDA addr` = 13T
Bit / Nibble / Byte / Word	1 bit / 4 bits / 8 bits / word-length group	—

Addressing capacity rule: an *n*-line address bus addresses **2^n** locations.

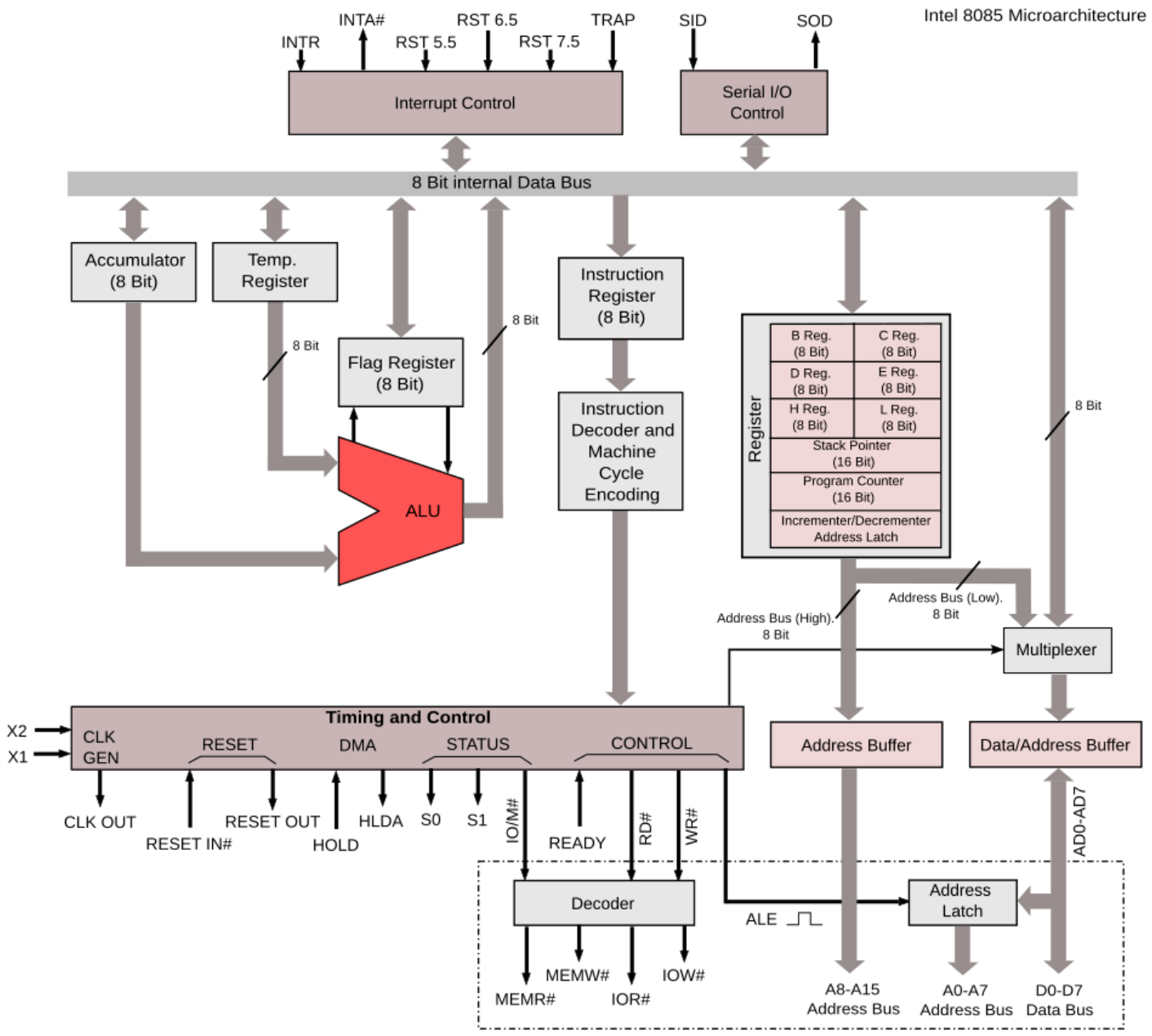
16 lines → 64 KB (8085); 20 lines → 1 MB (8086); 24 lines → 16 MB (80286).

5.1.3 Evolution snapshot (exam one-liners)

Processor	Year	Bits	Address bus	Note
Intel 4004	1971	4	12-bit	**First microprocessor**
Intel 8008	1972	8	14-bit	First 8-bit µP
Intel 8080	1974	8	16-bit	Needed +12 V, −5 V, +5 V
Intel 8085	1976	8	16-bit	Single **+5 V** supply, on-chip clock, serial I/O
Intel 8086	1978	16	20-bit	First x86; segmentation, pipelining
Intel 8088	1979	16 (8-bit ext. data bus)	20-bit	Used in IBM PC
80286 / 386 / 486 / Pentium	1982→	16/32	24/32-bit	Protected mode, cache, superscalar

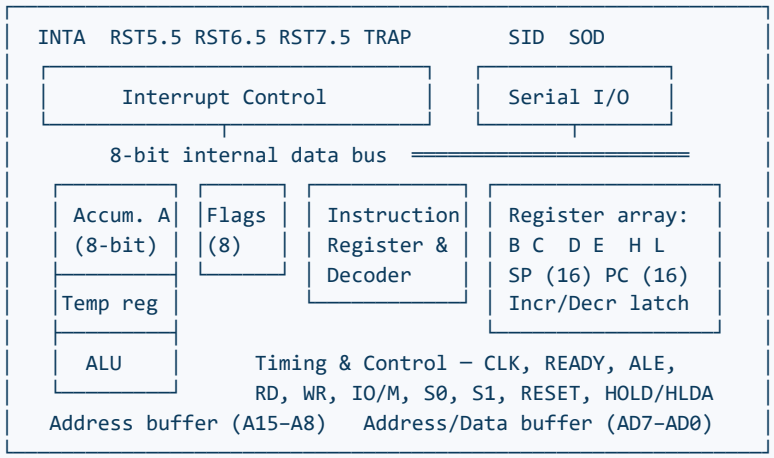
RISC vs CISC (quick): CISC (8085, 8086, x86) — many complex variable-length instructions, microprogrammed control. RISC (ARM, AVR, PIC newer cores) — few fixed-length instructions, load–store architecture, hardwired control, one instruction ≈ one cycle.

5.2 Intel 8085 — Architecture (Block Diagram)



Intel 8085 internal architecture.

The 8085 is an **8-bit, 40-pin DIP, NMOS** microprocessor running on a **single +5 V** supply. Its functional blocks:



5.2.1 ALU

Performs 8-bit **arithmetic** (add, subtract, increment, decrement, decimal adjust) and **logic** (AND, OR, XOR, complement, compare, rotate). Inputs: **Accumulator** and **Temporary register**; result normally goes back to the accumulator; **flags** are updated.

5.2.2 Register set

Register	Size	Role
A (Accumulator)	8-bit	One operand and destination of most ALU operations; used for I/O (`IN` / `OUT`)
B, C, D, E, H, L	8-bit each	General-purpose; usable in pairs **BC, DE, HL** as 16-bit registers
HL	16-bit pair	Default **memory pointer** — the operand "M" means the byte at address (HL)
PC (Program Counter)	16-bit	Holds address of the **next instruction byte** to be fetched
SP (Stack Pointer)	16-bit	Holds address of the **top of stack**; stack grows **downward** (toward lower addresses)
Instruction Register	8-bit	Holds the opcode currently being decoded (not programmer-accessible)
Temp / W / Z	8-bit	Internal scratch registers (not programmer-accessible)

5.2.3 Flag register (PSW low byte)

```
Bit:   D7   D6   D5   D4   D3   D2   D1   D0
Flag:  S    Z    x    AC   x    P    x    CY
```

Only **5 flags** are defined (D5, D3, D1 unused):

Flag	Name	Set (=1) when…
S	Sign	Bit D7 of the result is 1 (result negative in 2's-complement sense)
Z	Zero	Result is exactly 00H
AC	Auxiliary Carry	A carry propagates from bit D3 to bit D4 (used by `DAA` for BCD)
P	Parity	Result has an **even number of 1s** (even parity → P = 1)
CY	Carry	An arithmetic **carry out of D7** (add) or a **borrow** (subtract) occurs; also affected by rotate-through-carry instructions

> **Exam rules:** `INR` / `DCR` affect **all flags except CY**. Logical `ANA` sets CY = 0 and **AC = 1**; `ORA` / `XRA` set CY = 0 and AC = 0. `CMP` / `SUB` set CY = 1 when A < operand (borrow). Data-transfer instructions (`MOV` , `MVI` , `LXI` , `LDA` …) affect **no flags**.

5.2.4 Multiplexed bus and demultiplexing with ALE

To fit in 40 pins, the **low-order address byte A7–A0 shares pins with the data bus D7–D0**, called **AD7–AD0**.

- During **T1** of every machine cycle the pins carry the address low byte and **ALE (Address Latch Enable)** goes **high**.
- The falling edge of ALE clocks an external latch (**74LS373/8212**) that captures A7–A0.
- From T2 onward the same pins carry **data**.

A15–A8 are on dedicated pins throughout the cycle.

5.2.5 Control & status signals

Signal	Type	Meaning
ALE	Out	High pulse in T1; latch low address
RD̄	Out, active low	Read strobe — data valid on bus while low
WR̄	Out, active low	Write strobe
IO/M̄	Out	**1 = I/O operation, 0 = memory operation**
S1, S0	Out	Machine-cycle status (see table)
READY	In	Low ⇒ 8085 inserts **wait states** (slow memory/peripheral)
HOLD / HLDA	In / Out	DMA bus request / bus grant acknowledgement
RESET IN̄ / RESET OUT	In / Out	Reset: PC ← 0000H, buses tristated / reset for peripherals
CLK OUT	Out	System clock (crystal/2) for peripherals
X1, X2	In	Crystal connections

Status combinations:

IO/M̄	S1	S0	Machine cycle
0	1	1	**Opcode fetch**
0	1	0	Memory read
0	0	1	Memory write
1	1	0	I/O read
1	0	1	I/O write
1	1	1	Interrupt acknowledge
—	0	0	Halt

5.2.6 8085 pin summary (40-pin DIP)

- **Address:** A15–A8 (pins 21–28), AD7–AD0 (pins 12–19)
- **Control:** ALE(30), RD̄(32), WR̄(31), IO/M̄(34), S0(29), S1(33), READY(35)
- **Interrupts:** TRAP(6), RST 7.5(7), RST 6.5(8), RST 5.5(9), INTR(10), INTĀ(11)
- **Serial:** SID(5), SOD(4)
- **DMA:** HOLD(39), HLDA(38)
- **Clock/Reset:** X1(1), X2(2), CLK OUT(37), RESET IN̄(36), RESET OUT(3)
- **Power:** Vcc +5 V(40), Vss(20)

5.3 Instruction Cycle, Machine Cycles & T-States

Instruction cycle = Fetch + Decode + Execute. Each instruction takes **1 to 5 machine cycles**; each machine cycle takes **3 to 6 T-states**.

Machine cycle	T-states	Notes
Opcode fetch (OF)	**4T** (6T for `CALL`-class first cycle)	Only cycle with 4T minimum: T4 used for decoding
Memory read (MR)	3T	Fetch operand/address bytes
Memory write (MW)	3T	Store to memory
I/O read (IOR)	3T	`IN port`
I/O write (IOW)	3T	`OUT port`
Interrupt acknowledge (INA)	6T	For INTR response
Bus idle (BI)	2–3T	e.g. internal cycles of `DAD`

Timings of common instructions (memorise):

Instruction	Bytes	Machine cycles	T-states
`MOV r, r`	1	1 (OF)	**4T**
`MOV r, M` / `MOV M, r`	1	2 (OF + MR/MW)	**7T**
`MVI r, data`	2	2 (OF + MR)	**7T**
`MVI M, data`	2	3	10T
`LXI rp, data16`	3	3 (OF + 2 MR)	**10T**
`LDA addr`	3	4 (OF + 2 MR + MR)	**13T**
`STA addr`	3	4 (OF + 2 MR + MW)	**13T**
`ADD r`	1	1	4T
`ADD M`	1	2	7T
`INX rp` / `DCX rp`	1	1 (6T OF)	**6T**
`DAD rp`	1	3 (OF + 2 BI)	**10T**
`JMP addr`	3	3	**10T**
Conditional jump	3	2/3	**7T (false) / 10T (true)**
`CALL addr`	3	5	**18T**
`RET`	1	3	**10T**
`PUSH rp`	1	3	**12T**
`POP rp`	1	3	**10T**
`RST n`	1	3	12T
`XCHG`	1	1	4T
`NOP`	1	1	4T
`HLT`	1	1+	5T

Opcode-fetch timing (4T):

T1 — ALE high, AD bus = A7–A0, A15–A8 out, IO/$\bar{M}$=0 S1=1 S0=1.

T2 — $\bar{RD}$ low, memory places opcode on bus.

T3 — opcode latched into IR, $\bar{RD}$ high.

T4 — opcode decoded internally (bus idle).

5.4 Addressing Modes of the 8085

Mode	Operand location	Examples
Immediate	Inside the instruction itself	`MVI A, 32H` ; `ADI 47H` ; `LXI H, 2500H`
Register	In a CPU register	`MOV A, B` ; `ADD C` ; `XCHG`
Direct	16-bit address given in the instruction	`LDA 2050H` ; `STA 3000H` ; `IN 01H`
Register indirect	Address held in a register pair (usually HL)	`MOV A, M` ; `LDAX B` ; `STAX D`
Implied / Implicit	Operand implied by the opcode (usually A)	`CMA` ; `RLC` ; `DAA` ; `EI` ; `NOP`

> **Note:** The 8085 has **no indexed or relative addressing** (those appear in 8086/8051).

5.5 8085 Instruction Set — Group by Group

Total **246 opcodes / 74 instruction types**. Instruction lengths: 1-byte, 2-byte (opcode + data/port), 3-byte (opcode + 16-bit address/data).

5.5.1 Data transfer group (no flags affected)

Instruction	Action
`MOV rd, rs`	rd ← rs
`MOV r, M` / `MOV M, r`	register ↔ memory (HL)
`MVI r, d8` / `MVI M, d8`	Load immediate byte
`LXI rp, d16`	Load register pair immediate (BC/DE/HL/SP)
`LDA a16` / `STA a16`	A ↔ memory[direct address]
`LHLD a16` / `SHLD a16`	HL ↔ two memory bytes (L from a16, H from a16+1)
`LDAX B/D` / `STAX B/D`	A ↔ memory[(BC) or (DE)]
`XCHG`	HL ↔ DE
`IN p8` / `OUT p8`	A ↔ I/O port

5.5.2 Arithmetic group

Instruction	Action	Flags
`ADD r/M`	A ← A + operand	all
`ADI d8`	A ← A + immediate	all
`ADC r/M` / `ACI d8`	A ← A + operand + CY	all
`SUB r/M` / `SUI d8`	A ← A − operand	all (CY = borrow)
`SBB r/M` / `SBI d8`	A ← A − operand − CY	all
`INR r/M`	operand + 1	**all except CY**
`DCR r/M`	operand − 1	**all except CY**
`INX rp` / `DCX rp`	16-bit ±1	**no flags**
`DAD rp`	HL ← HL + rp	**only CY**
`DAA`	Decimal-adjust A after BCD addition	all

DAA rule: if lower nibble > 9 or AC = 1 → add 06H; if upper nibble > 9 or CY = 1 → add 60H.

5.5.3 Logical group

Instruction	Action	CY, AC after
`ANA r/M`, `ANI d8`	A ← A AND op	CY = 0, **AC = 1**
`ORA r/M`, `ORI d8`	A ← A OR op	CY = 0, AC = 0
`XRA r/M`, `XRI d8`	A ← A XOR op	CY = 0, AC = 0
`CMA`	A ← $\bar{A}$ (complement)	**no flags**
`CMC` / `STC`	CY ← $\bar{CY}$ / CY ← 1	only CY
`CMP r/M`, `CPI d8`	A − op (result discarded)	A < op ⇒ CY=1; A = op ⇒ Z=1; A > op ⇒ CY=0, Z=0
`RLC`	Rotate A left; D7 → D0 **and** → CY	only CY
`RRC`	Rotate A right; D0 → D7 and → CY	only CY
`RAL`	Rotate A left **through carry** (CY → D0, D7 → CY)	only CY
`RAR`	Rotate A right through carry	only CY

> `XRA A` is the standard trick to **clear the accumulator and CY** in one byte (also sets Z = 1, P = 1).

5.5.4 Branch group

Instruction	Condition
`JMP a16`	Unconditional
`JC` / `JNC`	CY = 1 / CY = 0
`JZ` / `JNZ`	Z = 1 / Z = 0
`JP` / `JM`	S = 0 (plus) / S = 1 (minus)
`JPE` / `JPO`	P = 1 (even) / P = 0 (odd)
`CALL a16` (+ conditional CC, CNC, CZ, CNZ, CP, CM, CPE, CPO)	Push PC on stack, jump
`RET` (+ conditional RC, RNC, RZ, RNZ, RP, RM, RPE, RPO)	Pop PC from stack
`PCHL`	PC ← HL (computed jump)
`RST 0…7`	1-byte call to vector **n × 8** (RST 5 → 0028H, RST 7 → 0038H)

5.5.5 Stack & machine-control group

Instruction	Action
`PUSH B/D/H/PSW`	(SP−1) ← high byte, (SP−2) ← low byte, SP ← SP − 2
`POP B/D/H/PSW`	low ← (SP), high ← (SP+1), SP ← SP + 2
`XTHL`	Exchange HL with top of stack
`SPHL`	SP ← HL
`EI` / `DI`	Enable / disable interrupts (set/clear INTE flip-flop)
`SIM`	**Set Interrupt Mask** + send serial output bit (uses A)
`RIM`	**Read Interrupt Mask** + read serial input bit (into A)
`HLT`	Halt CPU until interrupt or reset
`NOP`	No operation (4T delay / patching)

PSW = A (high byte) + Flags (low byte) when pushed/popped.

5.6 Assembly Language Programming — Worked Programs

Program 1 — 8-bit addition of two memory bytes

```
; Add bytes at 2050H and 2051H, store sum at 2052H, carry at 2053H
        LXI H, 2050H    ; HL points to first number
        MOV A, M        ; A <- (2050H)
        INX H           ; HL = 2051H
        MOV B, M        ; B <- (2051H)
        MVI C, 00H      ; carry counter = 0
        ADD B           ; A <- A + B
        JNC SKIP        ; no carry? skip
        INR C           ; record carry
SKIP:   INX H           ; HL = 2052H
        MOV M, A        ; store sum
        INX H
        MOV M, C        ; store carry
        HLT
```

Program 2 — 8-bit subtraction (with borrow indication)

```
; (2050H) - (2051H) -> 2052H ; CY = 1 indicates a borrow
        LDA 2050H       ; A <- minuend
        MOV B, A        ; save minuend in B
        LDA 2051H       ; A <- subtrahend
        MOV C, A        ; save subtrahend in C
        MOV A, B        ; A <- minuend
        SUB C           ; A <- minuend - subtrahend ; CY=1 if borrow
        STA 2052H       ; store difference
        HLT
```

Program 3 — Largest number in an array

```
; Array of N bytes starting at 2501H, count N at 2500H; result to 2600H
        LXI H, 2500H
        MOV C, M        ; C = count
        INX H
        MOV A, M        ; A = first element (current max)
        DCR C
LOOP:   INX H
        CMP M           ; compare A with next element
        JNC NEXT        ; A >= M ? keep A
        MOV A, M        ; else new max
NEXT:   DCR C
        JNZ LOOP
        STA 2600H
        HLT
```

Program 4 — Block transfer of 10 bytes (2001H → 3001H)

```
        MVI C, 0AH      ; counter = 10
        LXI H, 2001H    ; source pointer
        LXI D, 3001H    ; destination pointer
LOOP:   MOV A, M        ; read source byte
        STAX D          ; write to (DE)
        INX H
        INX D
        DCR C
        JNZ LOOP
        HLT
```

Program 5 — Count number of 1s in a byte

```
; Byte at 2050H; count of set bits -> 2051H
        LDA 2050H
        MVI B, 00H      ; ones counter
        MVI C, 08H      ; 8 bits to test
AGAIN:  RLC             ; rotate left, D7 -> CY
        JNC ZERO
        INR B
ZERO:   DCR C
        JNZ AGAIN
        MOV A, B
        STA 2051H
        HLT
```

Program 6 — Software delay loop (see Worked Example 5 for the calculation)

```
DELAY:  MVI C, FFH      ; 7T
LOOP:   DCR C           ; 4T
        JNZ LOOP        ; 10T taken / 7T last pass
        RET             ; 10T
```

Program 7 — 16-bit addition using DAD

```
; (2050H,2051H) + (2052H,2053H) -> (2054H,2055H)   [low byte first]
        LHLD 2050H      ; HL <- first 16-bit number
        XCHG            ; DE <- first number
        LHLD 2052H      ; HL <- second number
        DAD D           ; HL <- HL + DE, CY set on overflow
        SHLD 2054H
        HLT
```

5.7 8085 Interrupts

An **interrupt** is a hardware/software signal that suspends the main program and transfers control to a service routine (ISR).

5.7.1 The five hardware interrupts (priority order)

Interrupt	Priority	Trigger	Vector address	Maskable?
TRAP (RST 4.5)	1 (highest)	**Edge AND level** (rising edge, then held high)	**0024H**	**Non-maskable** (only disabled by reset/HLT states)
RST 7.5	2	**Positive (rising) edge-triggered**; latched by internal flip-flop	**003CH**	Maskable (SIM M7.5)
RST 6.5	3	**Level (high)** triggered	**0034H**	Maskable (SIM M6.5)
RST 5.5	4	**Level (high)** triggered	**002CH**	Maskable (SIM M5.5)
INTR	5 (lowest)	Level triggered	**No fixed vector** — external device supplies an instruction (usually `RST n` or `CALL`) during INTĀ cycle	Maskable by `DI`

Vector formula for software RST n: vector = **n × 0008H**. For the "half" interrupts: RST 5.5 → 5.5 × 8 = 44 = **002CH**, RST 6.5 → 52 = **0034H**, RST 7.5 → 60 = **003CH**, TRAP (4.5) → 36 = **0024H**.

> **PYQ fact:** RST 7.5 is the **only purely positive-edge-triggered** interrupt of the 8085. Because it is latched internally, a momentary pulse is remembered until served or reset (via SIM bit R7.5 or RESET).

5.7.2 SIM instruction format (contents of A before `SIM`)

```
D7    D6    D5    D4    D3    D2    D1    D0
SOD   SDE   x     R7.5  MSE   M7.5  M6.5  M5.5
```

- **M7.5/M6.5/M5.5 = 1 masks (disables)** that interrupt; effective only if **MSE = 1**.
- **R7.5 = 1** resets the RST 7.5 latch.
- **SDE = 1** enables serial output; **SOD** bit is then sent out on the SOD pin.

Example: enable only RST 6.5 → mask 7.5 & 5.5 → A = 0000 1101_2 = **0DH**, then `SIM` (after `EI`).

5.7.3 RIM instruction format (contents of A after `RIM`)

```
D7    D6    D5    D4    D3    D2    D1    D0
SID   I7.5  I6.5  I5.5  IE    M7.5  M6.5  M5.5
```

- I-bits = pending interrupts; IE = interrupt-enable flip-flop state; M-bits = current masks; SID = serial input pin level.

5.7.4 Serial I/O — SID and SOD

The 8085 has **software-driven single-line serial I/O**:

- **SID** (Serial Input Data, pin 5) read into bit D7 of A by `RIM`.
- **SOD** (Serial Output Data, pin 4) written from bit D7 of A by `SIM` (with SDE = 1).

Bit-banged serial transfer (framing, baud timing) is done entirely in software delay loops.

5.8 Memory Interfacing & Address Decoding

5.8.1 Steps

1. **Demultiplex** AD7–AD0 with ALE + 74LS373 → full 16-bit address A15–A0.
2. Generate **MEM$\bar{R}$ / MEM$\bar{W}$** from R$\bar{D}$, W$\bar{R}$ and IO/$\bar{M}$ (e.g., MEM$\bar{R}$ = R$\bar{D}$ OR IO/M).
3. **Address decoding:** low address lines go to the chip's address pins; the remaining high lines drive a decoder (74LS138) or logic gates to produce **chip select (C$\bar{S}$)**.

Lines needed by a chip: a 2^k-byte chip uses **k** address lines.

1 KB → 10 lines (A9–A0); 2 KB → 11; 4 KB → 12 (A11–A0); 8 KB → 13.

5.8.2 Worked memory map

Interface a 4 KB EPROM at 0000H and a 2 KB RAM at 3000H (see Worked Examples 6–7 for the decoding maths):

Device	Size	Address lines used	Decoded lines	Range
EPROM	4 KB	A11–A0	A15 A14 A13 A12 = 0 0 0 0	**0000H–0FFFH**
RAM	2 KB	A10–A0	A15...A11 = 0 0 1 1 0	**3000H–37FFH**

Range check: 4 KB = 1000H bytes → 0000H + 0FFFH ✓; 2 KB = 0800H bytes → 3000H + 07FFH = 37FFH ✓.

Absolute vs partial decoding: absolute decoding uses **all** unused high lines (unique map, more hardware); partial decoding ignores some lines → the device responds at **multiple (fold-back/shadow) ranges** — cheaper, used in small systems.

5.8.3 I/O interfacing: two schemes

Feature	I/O-mapped I/O (peripheral-mapped)	Memory-mapped I/O
Instructions	Only `IN p8`, `OUT p8`	All memory instructions (`MOV M`, `LDA`, `ADD M` ...)
Address size	8-bit → **256 input + 256 output ports**	16-bit (ports occupy memory space)
IO/$\bar{M}$ during access	1	0
Decoding hardware	Simpler (8 lines)	Full 16-line decode
Data transfer	Only via accumulator	Between memory-port and any register; ALU ops possible
Memory space	Fully preserved (64 KB)	Reduced by ports

5.9 Peripheral (Support) Chips

5.8.1 Intel 8255 — Programmable Peripheral Interface (PPI)

- 40-pin chip providing **24 programmable I/O lines**: **Port A**, **Port B**, **Port C** (C-upper CU + C-lower CL, 4 bits each).
- Two internal groups: **Group A** = PA + CU; **Group B** = PB + CL.
- Selected via A1 A0: 00 = Port A, 01 = Port B, 10 = Port C, 11 = **Control register**.

Operating modes

Mode	Description
Mode 0	Simple/basic I/O — no handshake; all 3 ports as plain input or output
Mode 1	**Strobed (handshake) I/O** — Ports A & B with Port C lines as handshake (STB̄, IBF, INTR / OBF̄, ACK̄)
Mode 2	**Bidirectional bus** on Port A only (with 5 Port C handshake lines); Port B may be Mode 0/1
BSR mode	Bit Set/Reset of individual **Port C** bits (control word D7 = 0)

I/O-mode control word (D7 = 1):

```
D7  D6 D5   D4   D3    D2   D1   D0
1   MA1,MA0 PA   PCU   MB   PB   PCL
|   Group A mode |     Group B mode  |
(port bit = 1 -> input, 0 -> output; MA 00=mode0 01=mode1 1x=mode2; MB 0=mode0 1=mode1)
```

Worked example: Mode 0, **PA = output, PB = input, PCU = input, PCL = output**:

D7=1, D6 D5=00, D4(PA)=0, D3(PCU)=1, D2(MB)=0, D1(PB)=1, D0(PCL)=0 → 1000 1010_2 = **8AH**.

BSR example: set bit PC3 → 0000 0111_2 = **07H** (D7=0, bits D3–D1 = 011 select PC3, D0 = 1 = set).

5.9.2 Intel 8253/8254 — Programmable Interval Timer

- **Three independent 16-bit down-counters** (Counter 0, 1, 2), each with CLK, GATE, OUT pins; max clock 8254 ≈ 8–10 MHz (8253 ≈ 2.6 MHz).
- **Six modes:** 0 = interrupt on terminal count; 1 = programmable one-shot; 2 = **rate generator (divide-by-N)**; 3 = **square-wave generator**; 4 = software-triggered strobe; 5 = hardware-triggered strobe.
- Control word selects counter, read/load format (LSB/MSB), mode, and binary/BCD counting.
- *Frequency rule:* OUT frequency = CLK frequency ÷ N (mode 2/3).

5.9.3 Intel 8257 — DMA Controller (also study 8237 conceptually)

DMA (Direct Memory Access) = high-speed data transfer between peripheral (disk, ADC) and memory **without CPU involvement in each byte**.

Handshake sequence (exam favourite):

1. Peripheral raises **DRQ (DMA request)** to the 8257 channel.
2. 8257 asserts **HRQ → HOLD** input of the 8085.
3. 8085 finishes the **current machine cycle**, tristates its buses, and replies **HLDA** (Hold Acknowledge).
4. The 8257 becomes **bus master**: it drives the address bus (its channel address register contents are placed on the bus, upper byte via an external latch) and control lines, and issues **DACK̄ (DMA acknowledge) to the requesting peripheral** — DACK works as the peripheral's chip-select, telling it the bus is granted and the memory read/write cycle is beginning.
5. Bytes transfer directly memory ↔ peripheral; the channel's 14-bit terminal counter decrements; on terminal count **TC** goes high.
6. DRQ removed → 8257 drops HRQ → 8085 regains the bus.

> **PYQ timing point: DACK is issued only after HLDA is received and the DMA controller has taken over and placed the address on the bus** — i.e., at the start of each DMA transfer cycle for that channel, not at the instant of DRQ.

- 8257: **4 channels** (CH0–CH3), each with 16-bit address register + 14-bit counter → up to **16 KB per programmed block**.
- **Transfer modes: Cycle stealing** — one byte per DMA cycle, bus returned to CPU between bytes (CPU slowed slightly); **Burst/Block mode** — whole block moved while CPU is held (fastest); (8237 also offers demand mode).

5.9.4 Intel 8259 — Programmable Interrupt Controller (PIC)

- Expands 8085/8086 interrupt capability: **8 vectored priority interrupt inputs (IR0–IR7)**; cascadable (1 master + 8 slaves) to **64 levels**.
- Contains IRR (request register), ISR (in-service), IMR (mask), priority resolver.
- Programmed with ICWs (initialisation) and OCWs (operation); supports fully-nested, rotating, and specific-mask priority schemes.
- Works with INTR/INTĀ: on INTĀ it supplies a `CALL` opcode + vector address (8085 mode) or vector type (8086 mode).

5.9.5 Intel 8279 — Programmable Keyboard/Display Interface

- Scans up to a **64-key matrix (8 × 8)** with 2-key lockout / N-key rollover; debouncing built in; 8-character **FIFO** for key codes (interrupts CPU via IRQ).
- Drives a multiplexed display of up to **16 digits** (7-segment) from internal 16 × 8 display RAM — relieves the CPU of refresh duty.

5.9.6 Other classic peripherals (one-liners)

- **8251 USART** — programmable serial communication (sync/async).
- **8155/8156** — 256-byte RAM + 22 I/O lines + 14-bit timer (companion to 8085 multiplexed bus).
- **8212/74LS373** — 8-bit latch for address demultiplexing.
- **8284 / 8288** — clock generator / bus controller for 8086.

5.10 Other Microprocessors — the 8086 in One Page

Intel 8086 (1978): first **16-bit** x86 processor.

Feature	Detail
Data bus	16-bit
Address bus	**20-bit → 1 MB** physical memory
Clock	5/8/10 MHz versions
Architecture	Split into **BIU** (Bus Interface Unit: segment registers, IP, 6-byte instruction **queue** → pipelining) and **EU** (Execution Unit: ALU, general registers, flags)
Registers	AX BX CX DX (each splittable AH/AL…), SP BP SI DI, segment CS DS SS ES, IP, 16-bit FLAGS (9 flags: CF PF AF ZF SF TF IF DF OF)
I/O space	64 K ports (16-bit port address)
Modes	Minimum (single processor) / Maximum (co-processor 8087, 8288 bus controller)

Memory segmentation: memory is accessed through 64 KB **segments**. Physical address (20-bit) is formed as:

```
Physical address = Segment register × 10H + Offset
Example: CS = 2000H, IP = 1234H  →  20000H + 1234H = 21234H
```

Segmentation allows relocatable code and separates code/data/stack.

8085 vs 8086 key differences

Point	8085	8086
Word length	8-bit	16-bit
Address bus / memory	16-bit / 64 KB	20-bit / 1 MB
Pipelining	None	6-byte prefetch queue
Multiply/Divide instructions	**No** (software only)	**Yes (MUL, DIV)**
Segmentation	No	Yes
Flags	5	9
Clock (typ.)	3 MHz	5–10 MHz
Duplicated in 8088	—	Same core, 8-bit external bus

5.11 Microprocessor vs Microcontroller

Feature	Microprocessor (8085/8086)	Microcontroller (8051)
Definition	CPU only on chip	**Computer-on-a-chip**: CPU + ROM + RAM + I/O + timers + serial port
External parts	Needs external memory & I/O chips	Mostly self-contained
Application	General-purpose computing	Embedded/dedicated control (washing machines, meters, automotive)
Instruction focus	Data processing	**Bit-manipulation** (Boolean processor) & I/O intensive
Power / cost / size	Higher	Low power, low cost, compact
Bus access	Address/data buses brought out	Buses mostly internal (P0/P2 can extend)
Registers	Few, operations register-based	More on-chip RAM; register banks
Examples	8085, 8086, Pentium, Core	8051, PIC16F, AVR ATmega, ARM Cortex-M MCUs

5.12 The 8051 Microcontroller

The Intel **8051 (MCS-51 family, 1980)** is the world's most exam-quoted 8-bit microcontroller.

5.12.1 Headline specifications (memorise)

Feature	Value
CPU	8-bit
On-chip **ROM**	**4 KB** (8031 = 0 KB, 8751 = 4 KB EPROM, 8052 = 8 KB)
On-chip **RAM**	**128 bytes** (8052 = 256 B)
Register banks	**Four banks** (Bank 0–3), each with **R0–R7**
I/O ports	**Four 8-bit ports P0–P3 = 32 I/O lines**
Timers	**Two 16-bit** timers/counters T0, T1 (8052 adds T2)
Interrupt sources	**Five** (2 external + 2 timer + 1 serial)
Serial port	Full-duplex UART (SBUF, 4 modes)
External memory	Up to 64 KB program + 64 KB data (16-bit address via P0/P2 + ALE)
Machine cycle	**12 clock periods** (e.g., 12 MHz crystal → 1 µs machine cycle)
Pins / package	40-pin DIP
Special pins	ALE, PSEN̄ (program store enable — external code fetch), EĀ (external access: 0 = ignore internal ROM), RST (active **high**, ≥ 2 machine cycles)

5.12.2 Internal RAM map (128 bytes: 00H–7FH)

Address	Contents
00H–1FH	**Four register banks × 8 bytes** (Bank0: 00–07, Bank1: 08–0F, Bank2: 10–17, Bank3: 18–1F)
20H–2FH	**Bit-addressable area** — 16 bytes = **128 individually addressable bits** (bit addresses 00H–7FH)
30H–7FH	General-purpose scratchpad / stack area (stack grows **upward**; SP reset value = **07H**)
80H–FFH (direct)	**SFR space** (ports, control registers)

Bank selection — PSW bits RS1, RS0 (PYQ fact):

RS1	RS0	Active bank	Addresses
0	0	Bank 0 (default after reset)	00H–07H
0	1	Bank 1	08H–0FH
1	0	Bank 2	10H–17H
1	1	Bank 3	18H–1FH

PSW format: `CY AC F0 RS1 RS0 OV – P` (P = parity of accumulator, updated every cycle; OV = overflow for signed arithmetic).

5.12.3 Important SFRs

A (E0H), B (F0H), PSW (D0H), SP (81H), DPTR = DPH:DPL (83H:82H), P0 (80H), P1 (90H), P2 (A0H), P3 (B0H), TMOD (89H), TCON (88H), TH0/TL0 (8CH/8AH), TH1/TL1 (8DH/8BH), SCON (98H), SBUF (99H), IE (A8H), IP (B8H), PCON (87H). SFRs whose address ends in 0H or 8H are **bit-addressable**.

5.12.4 Ports P0–P3

Port	Alternate function
P0	Multiplexed **AD7–AD0** for external memory; **open-drain** — needs external pull-ups for plain I/O
P1	Plain I/O only (8052: P1.0 = T2, P1.1 = T2EX)
P2	High address byte **A15–A8** for external memory
P3	P3.0 RXD, P3.1 TXD, P3.2 $\overline{\text{INT0}}$, P3.3 $\overline{\text{INT1}}$, P3.4 T0, P3.5 T1, P3.6 $\overline{\text{WR}}$, P3.7 $\overline{\text{RD}}$

To use a port pin as **input**, first write **1** to it (quasi-bidirectional ports).

5.12.5 Timers T0 & T1 — TMOD and TCON

TMOD (89H): upper nibble = Timer 1, lower = Timer 0. Each nibble = `GATE C/T M1 M0`.

- **C/$\overline{\text{T}}$ = 0** → timer (counts machine cycles = XTAL/12); **= 1** → counter (counts pulses on T0/T1 pin).
- **GATE = 1** → run only while INT$\overline{\text{x}}$ pin is high (hardware gating).

M1 M0	Mode	Description
0 0	**Mode 0**	13-bit timer (8192 counts) — legacy
0 1	**Mode 1**	**16-bit timer** (65536 counts) — most used
1 0	**Mode 2**	**8-bit auto-reload** (TH reloads TL) — baud-rate generation
1 1	Mode 3	Split timer (T0 as two 8-bit timers)

TCON (88H, bit-addressable): `TF1 TR1 TF0 TR0 IE1 IT1 IE0 IT0` — TRx starts the timer, TFx = overflow flag; ITx = 1 selects **edge**-trigger for INT$\overline{\text{x}}$.

Count/overflow rule (mode 1): delay = (65536 − initial count) × machine-cycle time. See Worked Example 9.

5.12.6 Interrupts — five sources

Source	Flag	Vector	Default priority
External $\overline{\text{INT0}}$	IE0	**0003H**	1 (highest)
Timer 0 overflow	TF0	**000BH**	2
External $\overline{\text{INT1}}$	IE1	**0013H**	3
Timer 1 overflow	TF1	**001BH**	4
Serial (RI or TI)	RI/TI	**0023H**	5 (lowest)

(Reset vector = 0000H; vectors are 8 bytes apart.)

IE register: `EA – ET2 ES ET1 EX1 ET0 EX0` — EA must be 1 to enable anything. **IP register** raises chosen sources to high priority.

5.12.7 Serial port (SCON modes)

Mode	SM0 SM1	Description	Baud
0	00	Shift register (synchronous)	XTAL/12
1	01	**8-bit UART** (most used)	Variable — Timer 1 mode 2
2	10	9-bit UART	XTAL/32 or /64
3	11	9-bit UART	Variable

Baud (mode 1, SMOD=0) = XTAL / (12 × 32 × (256 − TH1)). With 11.0592 MHz and TH1 = FDH (−3): 11 059 200/(384 × 3) = **9600 baud**.

5.12.8 8051 assembly flavour — quick examples

```
; Toggle all bits of Port 1 forever
BACK:  MOV  A, #55H     ; 0101 0101
       MOV  P1, A
       ACALL DELAY
       CPL  A           ; complement -> AAH
       MOV  P1, A
       ACALL DELAY
       SJMP BACK
```

```
; Timer 0, mode 1: create one 50 ms tick @ 12 MHz (count = 15536 = 3CB0H)
       MOV  TMOD, #01H  ; T0 mode 1
       MOV  TH0, #3CH
       MOV  TL0, #0B0H
       SETB TR0         ; start
WAIT:  JNB  TF0, WAIT   ; poll overflow
       CLR  TR0
       CLR  TF0
```

Addressing modes of 8051: immediate (`#`), register (Rn), direct (RAM/SFR address), register-indirect (`@R0/@R1` , `@DPTR` via MOVX), indexed (`MOVC A,@A+DPTR` — table lookup), plus rich **bit addressing** (`SETB` , `CLR` , `JB` , `JNB`).

5.12.9 Modern microcontrollers (one-liners)

- **AVR (Atmel/Microchip ATmega328 — Arduino Uno):** 8-bit **RISC**, Harvard architecture, most instructions in 1 clock, flash program memory.
- **PIC (Microchip PIC16/18):** 8-bit RISC Harvard, banked RAM, very popular in low-cost control.
- **ARM Cortex-M (STM32, LPC, nRF): 32-bit RISC** licensed core, Thumb-2 instruction set, NVIC interrupt controller — dominant modern embedded platform.
- **MSP430 (TI):** 16-bit ultra-low-power RISC MCU.
- **Harvard vs Von Neumann:** Harvard = separate program & data memory/buses (8051, AVR, PIC); Von Neumann = shared memory/bus (8085, 8086).

5.13 Worked Examples (10)

WE-1 (Hex addition + all flags). `MVI A, 9CH` → `MVI B, 64H` → `ADD B` . Find A and flags.

9CH + 64H: C+4 = 10H → nibble 0, AC = 1. 9+6+1 = 10H → result high nibble 0, carry out = 1.

A = 00H; CY = 1, AC = 1, Z = 1, S = 0, P = 1 (zero has even number of 1s). ✓ (9C = 156, 64H = 100, 156+100 = 256 = 100H.)

WE-2 (Subtraction/borrow). A = 45H, `SUI 62H`.

45H − 62H = 45 + (2's comp of 62 = 9EH) = E3H with no carry out → **CY = 1 (borrow)**. A = **E3H**, S = 1, Z = 0, P: E3 = 1110 0011 → six 1s → **P = 1**. (Check decimal: 69 − 98 = −29; E3H = 227 = 256 − 29 ✓.)

WE-3 (Logic + rotate). A = C5H. After `ANI 0FH` → A = 05H (CY=0, AC=1). Then `RLC` → 0000 1010 = **0AH**, CY = 0. Then `RRC` twice on 0AH: 0AH → 05H (CY=0) → 82H (CY=1, since D0 of 05H = 1 rotates into D7). Final **A = 82H, CY = 1**.

WE-4 (Program trace).

```
MVI A, 07H
RLC
MOV B, A
RLC
RLC
ADD B
```

07H → RLC → 0EH; B = 0EH; RLC → 1CH; RLC → 38H; 38H + 0EH = **46H**. (This computes 7 × 10: 7×2 + 7×8 = 70 = 46H ✓.)

WE-5 (Delay calculation). Crystal 6.144 MHz → clock 3.072 MHz → T = 325.5 ns. Loop:

```
MVI C, FFH   ; 7T (once)
LOOP: DCR C  ; 4T
JNZ LOOP     ; 10T (254 times) / 7T (last)
```

T-states = 7 + 255 × 4 + 254 × 10 + 7 = 7 + 1020 + 2540 + 7 = **3574T** → 3574 × 0.3255 µs ≈ **1.163 ms**. General inner-loop formula: ≈ 14 × N T-states (+ overhead), N = count.

WE-6 (Address decoding). A 74LS138 decoder driven by A15 A14 A13 (E-inputs tied active). Output Y5 selects an 8 KB RAM. Range?

Y5 → A15 A14 A13 = 1 0 1; chip uses A12–A0 (13 lines). Range = 1010 0000 0000 0000 to 1011 1111 1111 1111 = **A000H–BFFFH** (size BFFF−A000+1 = 2000H = 8 KB ✓).

WE-7 (Chip-select logic). A 1 KB EPROM must sit at 0800H–0BFFH. 1 KB → A9–A0 to the chip. Fixed high bits: 0800H = 0000 10xx …; so **A15 A14 A13 A12 A11 A10 = 0 0 0 0 1 0**. $\overline{CS}$ = NAND of ($\bar{A}15 \cdot \bar{A}14 \cdot \bar{A}13 \cdot \bar{A}12 \cdot A11 \cdot \bar{A}10$) — e.g., a 6-input gate arrangement. Verify top: 0000 1011 1111 1111 = 0BFFH ✓.

WE-8 (Interrupt vector). A device pulls RST 6.5 high while IE = 1 and M6.5 = 0. Where does the 8085 go? Vector = 6.5 × 8 = 52 = **0034H**. (The CPU internally executes a restart: pushes PC, jumps to 0034H.)

WE-9 (8051 timer count). Generate a **2 ms** delay with Timer 1, mode 1, XTAL = 12 MHz.

Machine cycle = 12/12 MHz = 1 µs → need 2000 counts. Initial value = 65536 − 2000 = 63536 = **F830H** → `MOV TH1,#0F8H`, `MOV TL1,#30H`. (Check: F830H = 63536; 65536 − 63536 = 2000 ✓.)

WE-10 (8051 max timer delay & 8255 word). (a) Max mode-1 delay @ 12 MHz = 65536 µs ≈ **65.54 ms** (TH=TL=00H). (b) 8255 control word for **all three ports output, mode 0**: 1000 0000 = **80H**; for **all input**: 1001 1011 = **9BH**.

5.14 PYQ Corner — 22-04-2025 CBT-2 (solved)

Q1. After executing the program below, the content of the accumulator is:

```
MVI B, 06H
MVI A, F2H
ADD B
```

(a) F6H (b) F8H (c) 08H (d) F2H

Solution. `MVI B,06H` loads B = 06H (no flags). `MVI A,F2H` loads A = F2H. `ADD B`: F2H + 06H → 2+6 = 8 (no aux carry), F+0 = F → **A = F8H**; CY = 0, Z = 0, S = 1 (D7=1), P: F8 = 1111 1000 has five 1s → P = 0. **Answer: (b) F8H.**

Q2. The number of register banks available in the 8051 microcontroller is:

(a) Two (b) Three (c) Four (d) Eight

Solution. Internal RAM 00H–1FH holds **four register banks** (Bank 0–3), each containing registers **R0–R7** (8 bytes per bank = 32 bytes total). The active bank is chosen by bits **RS1, RS0 of the PSW**; Bank 0 is default after reset. **Answer: (c) Four.**

Q3. Which 8085 interrupt is positive-edge-triggered?

(a) TRAP (b) RST 7.5 (c) RST 6.5 (d) INTR

Solution. RST 7.5 is triggered by a **rising (positive) edge** and is internally **latched** — the request is remembered even if the pin returns low, until serviced or cleared by SIM bit R7.5/reset. RST 6.5, RST 5.5 and INTR are level-triggered; TRAP needs **both** an edge and sustained level. **Answer: (b) RST 7.5.**

Q4. In a DMA transfer using the 8257, the DACK̄ signal is sent to the peripheral:

(a) immediately when the peripheral raises DRQ

(b) after the CPU acknowledges HOLD with HLDA and the DMA controller, as bus master, begins the transfer cycle for that channel

(c) only after the entire block is transferred

(d) simultaneously with the HOLD request

Solution. Sequence: DRQ → 8257 raises HRQ/HOLD → CPU completes its current machine cycle, floats the buses and returns **HLDA** → 8257 takes bus mastership, **places the memory address on the bus, and only then asserts DACK̄** to the requesting peripheral, which uses it as a chip-select for the actual read/write cycle. DACK therefore marks the **grant of the bus and the start of the DMA cycle**, not the request. **Answer: (b).**

5.15 Practice MCQs (25)

1. The number of address lines in the 8085 is:

(a) 8 (b) 16 (c) 20 (d) 40

2. The 8085 flag register does **not** contain:

(a) Sign (b) Overflow (c) Parity (d) Auxiliary carry

3. The instruction `MOV A, M` uses which addressing mode?

(a) Direct (b) Immediate (c) Register indirect (d) Implied

4. An opcode fetch machine cycle of the 8085 requires a minimum of:

(a) 3 T-states (b) 4 T-states (c) 5 T-states (d) 6 T-states

5. After `XRA A`, the flags are:

(a) Z=1, CY=1 (b) Z=1, CY=0, P=1 (c) Z=0, CY=0 (d) Z=1, P=0

6. ALE of the 8085 is used to:

(a) enable interrupts (b) latch the low-order address (c) strobe data writes (d) select I/O

7. The vector address of RST 5.5 is:

(a) 002CH (b) 0034H (c) 003CH (d) 0024H

8. Which interrupt of the 8085 is non-maskable?

(a) RST 7.5 (b) INTR (c) TRAP (d) RST 5.5

9. `PUSH B` changes the stack pointer by:

(a) +1 (b) +2 (c) −1 (d) −2

10. If A = 3FH, after `INR A` the flags AC and Z are:

(a) AC=1, Z=0 (b) AC=0, Z=0 (c) AC=1, Z=1 (d) AC=0, Z=1

11. The number of I/O ports addressable with 8085 `IN` / `OUT` instructions:

(a) 256 in + 256 out (b) 64 K (c) 128 (d) 65536 combined

12. In memory-mapped I/O:

(a) only IN/OUT can access ports (b) arithmetic can be done directly on port data (c) IO/$\bar{M}$ = 1 during access (d) 256 ports maximum

13. The 8255 control word 90H (mode 0) configures:

(a) PA in, PB out, PC out (b) PA out, PB in (c) all ports input (d) all ports output

14. The mode of the 8253/54 used as a square-wave generator is:

(a) Mode 0 (b) Mode 1 (c) Mode 3 (d) Mode 5

15. During DMA, the 8085 responds to HOLD with:

(a) INT$\bar{A}$ (b) HLDA (c) READY (d) ALE

16. Cycle-stealing DMA means:

(a) the CPU is halted for the full block (b) one byte is transferred per bus grant, interleaved with CPU cycles (c) transfer without DACK (d) transfer via the accumulator

17. The 8259 PIC, fully cascaded, supports:

(a) 8 interrupts (b) 16 (c) 64 (d) 256

18. Physical address for CS = 1000H, IP = 2000H in the 8086:

(a) 3000H (b) 12000H (c) 102000H (d) 21000H

19. The 8086 instruction queue in the BIU is:

(a) 4 bytes (b) 6 bytes (c) 8 bytes (d) 16 bytes

20. On-chip data RAM of the standard 8051 is:

(a) 64 B (b) 128 B (c) 256 B (d) 4 KB

21. With a 12 MHz crystal, one 8051 machine cycle takes:

(a) 83 ns (b) 0.5 µs (c) 1 µs (d) 12 µs

22. Which 8051 port is open-drain and needs external pull-ups for I/O use?

(a) P0 (b) P1 (c) P2 (d) P3

23. Timer mode 2 of the 8051 is:

(a) 13-bit (b) 16-bit (c) 8-bit auto-reload (d) split timer

24. The 8051 interrupt with the highest default priority is:

(a) Timer 0 (b) Serial (c) External INT0 (d) Timer 1

25. Register banks in the 8051 are selected by:

(a) bits in TCON (b) RS1, RS0 in PSW (c) bits in IE (d) the SP value

Answer Key with brief explanations

Q	Ans	Why
1	b	A15–A0 → 64 KB
2	b	8085 has S, Z, AC, P, CY only; overflow flag appears in 8086/8051 (OV)
3	c	Operand address in HL pair
4	b	4T (T4 = decode); some opcodes take a 6T fetch
5	b	A XOR A = 00H → Z=1, CY=0, AC=0, S=0, P=1 (even)
6	b	ALE clocks the external 74LS373 in T1
7	a	5.5 × 8 = 44 = 002CH
8	c	TRAP (RST 4.5) is non-maskable, vector 0024H
9	d	Two bytes pushed → SP ← SP − 2
10	a	3FH+1 = 40H: carry from D3→D4 ⇒ AC=1; result ≠ 0 ⇒ Z=0 (CY unchanged by INR)
11	a	8-bit port address, separate read/write spaces
12	b	Ports behave as memory bytes; any memory instruction works
13	a	1001 0000: D4=1 → PA input; D1=0 → PB output; D3=D0=0 → PC output
14	c	Mode 3 = square wave; Mode 2 = rate generator
15	b	HLDA acknowledges bus release
16	b	Byte-at-a-time interleave; burst mode holds the bus for the block
17	c	1 master + 8 slaves × 8 = 64 levels
18	b	1000H × 10H + 2000H = 10000H + 2000H = 12000H
19	b	6-byte prefetch queue (8088 has 4)
20	b	128 bytes (00–7FH); 8052 has 256 B
21	c	12 clocks per machine cycle → 12/12 MHz = 1 µs
22	a	P0 doubles as AD7–AD0; open-drain outputs
23	c	TH auto-reloads TL on overflow — used for baud rates
24	c	Order: INT0 > TF0 > INT1 > TF1 > Serial
25	b	RS1 RS0 = 00/01/10/11 → Bank 0/1/2/3

5.16 Rapid-Revision Sheet

8085 interrupt vector table

Interrupt	Trigger	Vector	Maskable
TRAP	edge + level	0024H	No
RST 7.5	**+ve edge (latched)**	003CH	Yes (SIM)
RST 6.5	level	0034H	Yes
RST 5.5	level	002CH	Yes
INTR	level	external (RST n/CALL)	Yes (DI)
RST n (software)	—	n × 8	—

Flag rules in one breath

- **CY**: carry out of D7 / borrow; unaffected by INR, DCR, INX, DCX; cleared by ANA/ORA/XRA; only flag touched by DAD, STC, CMC, rotates.
- **AC**: carry D3 → D4; ANA forces AC = 1.
- **Z**: result = 00H. **S**: copy of D7. **P**: 1 for even count of 1s.
- CMP/CPI: A < op → CY=1; A = op → Z=1; A > op → CY=0 & Z=0.

T-state quick table

Item	T
Opcode fetch	4 (or 6)
Memory/I-O read or write cycle	3
MOV r,r / ADD r / NOP	4
MVI r / MOV r,M / ADD M	7
LXI / JMP / DAD / RET / POP	10
PUSH / RST	12
LDA / STA	13
CALL	18
Conditional jump	7 fail / 10 pass

8085 vs 8086 vs 8051 comparison

Feature	8085	8086	8051
Type	8-bit µP	16-bit µP	8-bit µC
Address bus / memory	16-bit / 64 KB	20-bit / 1 MB	64 KB code + 64 KB data (ext.)
On-chip memory	None	None	4 KB ROM + 128 B RAM
Architecture	Von Neumann	Von Neumann (BIU+EU pipeline)	**Harvard**
Flags	5	9	PSW (CY AC F0 RS1 RS0 OV P)
Interrupt inputs	5 hardware	INTR + NMI (256 types)	5 sources, 2 priority levels
Timers on chip	None	None	Two 16-bit (T0, T1)
Multiply/Divide instr.	No	Yes	Yes (MUL AB, DIV AB)
Machine cycle	3–6 T-states	4 T-states basic bus cycle	**12 clock periods**
Register banks	No	No	**4 banks (R0–R7), RS1 RS0**
Serial port	SID/SOD (software)	External 8251	Built-in full-duplex UART
Typical use	Learning/legacy systems	PCs (x86 origin)	Embedded control

Peripheral chips in one line each

Chip	Role
8255	PPI — 24 I/O lines, modes 0/1/2 + BSR
8253/54	3 × 16-bit programmable timers, 6 modes
8257	4-channel DMA controller (DRQ/DACK, HRQ/HLDA)
8259	Priority interrupt controller, 8→64 levels
8279	Keyboard scan (64 keys) + display refresh (16 digits)
8251	USART serial communication
8155	RAM + I/O + timer companion for 8085

Last-minute one-liners

- First microprocessor = **Intel 4004** (1971, 4-bit).
- 8085: 40-pin, +5 V, 246 opcodes, 74 instructions, 6.144 MHz crystal → 3.072 MHz clock.
- ALE latches A7–A0; IO/$\bar{M}$ = 1 for I/O, 0 for memory.
- `XRA A` clears A and CY together.
- Stack grows downward in 8085 (upward in 8051; 8051 SP reset = 07H).
- 8051 vectors: 0003, 000B, 0013, 001B, 0023H (8 bytes apart).
- 9600-baud magic crystal for 8051 = **11.0592 MHz** with TH1 = FDH.
- DAC$\bar{K}$ = peripheral's chip-select during a DMA cycle, issued after HLDA/bus grant.

End of Chapter 5. Next: Chapter 6 as per syllabus sequence.